Tamás Bukits

Computer Vision Engineer & 3D Reconstruction Researcher

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PROFILE

Industrial PhD Researcher at Vicomtech working at the intersection of computer vision, neural rendering, and extended reality, designing AI-driven pipelines that transform real-world people, objects, and environments into immersive digital experiences. My research integrates multimodal sensing, semantic scene understanding, volumetric capture, 3D Gaussian Splatting, and digital twins, with a focus on developing scalable, human-centred solutions for next-generation immersive applications.

PROFESSIONAL EXPERIENCE

Researcher, Interactive Media — Feb. 2024 – Present

Vicomtech — San Sebastián, Spain

- Progressed from Computer Vision Research Intern to Research Assistant and Industrial PhD Researcher in the Digital Media Department.
- Develop AI-driven pipelines for reconstructing real-world scenes, people, and objects into immersive XR environments.
- Research volumetric capture and 3D Gaussian Splatting, including methods that reduce dependence on Structure-from-Motion (SfM) and COLMAP-based initialization.
- Integrate semantic segmentation and multimodal scene understanding into live and offline volumetric media workflows.
- Contribute to European and industrial R&D projects in immersive media, digital twins, healthcare, and human-centred XR.
- Publish and present research at international conferences as both first author and co-author.

Software Engineer — Jan. 2022 – Jan. 2024

SwiconGroup Zrt. — Budapest, Hungary

- Built backend and frontend functionality for contract-management and HR workflow products using Node.js, TypeScript, Angular and MongoDB.
- Developed features for a Java-based sales management application and collaborated across product and engineering teams.

EDUCATION

Industrial PhD in Communication Technologies and Systems — Oct. 2025 – Present

Polytechnic University of Madrid — Madrid, Spain

Thesis: *Bringing Real World Entities Into Immersive Environments*: Designing AI-driven pipelines that digitise real-world people, objects, and environments using multimodal sensing, 3D scene understanding, and neural rendering to create scalable, high-fidelity immersive experiences for extended reality.

Erasmus Mundus Joint MSc in Image Processing and Computer Vision — Sep. 2022 – Jun. 2024

Pázmány Péter Catholic University · Autonomous University of Madrid · University of Bordeaux — Hungary · Spain · France

Grade: 4.11/5.00

Thesis: *AI Enhancement of Volumetric Captures* — a Gaussian Splatting pipeline designed to reduce reliance on Structure-from-Motion and COLMAP.

BSc in Computer Science Engineering — Sep. 2017 – Jun. 2021

Budapest University of Technology and Economics — Budapest, Hungary

Grade: 4.49/5.00

Thesis: *Algorithms for Smoothing Polyhedra* — a framework for generating smoothly connected surfaces from control polyhedra.

RESEARCH PROJECTS

BAZKARIA — **Immersive telepresence for shared dining**

Human-centered XR research exploring remote dining, volumetric presence and interaction design in collaboration with Mugaritz.
Role: Developed a real-time pipeline for capturing, processing, transmitting and rendering 3D holograms represented as point clouds, from depth-camera acquisition to display on HDMI-enabled devices.

SCENARIST — **AI-powered synthetic data generation for safer public spaces**

End-to-end framework for creating photorealistic, automatically labelled synthetic datasets through 3D digitisation, Unreal Engine simulation and digital-twin technologies for AI training.
Role: Contributed to 3D scanning, reconstruction and digitisation of real-world people and objects, enabling the generation of realistic digital assets for large-scale synthetic data creation using Gaussian Splatting

AMPLIFY — **AI-generated immersive 3D memories**

Volumetric capture and evaluation of AI-based 3D reconstruction pipelines for immersive cultural experiences at the Gran Teatre del Liceu.
Role: Developed and evaluated 3D reconstruction pipelines for objects and people using generative AI and 3D Gaussian Splatting.

INSPIRE — **Volumetric digitization for XR training**

Volumetric capture and 3D reconstruction of people and environments to create realistic, reusable assets for immersive training applications.
Role: Developed 3D reconstruction and volumetric digitization workflows for people and environments using 3D Gaussian Splatting.

AUTOPILOT — **Digital surgical twins for robotic laparoscopy**

3D reconstruction from laparoscopic video and digital-twin technologies supporting surgical planning, spatial understanding and increased robotic autonomy.
Role: Contributed to laparoscopic 3D reconstruction and the development of digital surgical twins.

SYNERGIA — **Adaptive virtual environments powered by generative AI**

Human-centered XR combining multimodal sensing, generative AI, human digital twins and intelligent agents to personalize training, rehabilitation and skills development.
Role: Contributed to 3D reconstruction, human digital avatars and multimodal XR research.

PUBLICATIONS

A Sneak Peek into AI-Generated 3D Immersive Memories — ACM IMX 2026 · WiP
P. de Torres Coll, **T. Bukits**, J. Ruiz, I. Burguera Hidalgo, M. Zorrilla and S. Cabrero Barros.

The XR Table: Envisioning the Future of Remote Dining Experiences Using Immersive Telepresence — ACM DIS 2025
A. Santos-Torres, P. de Torres Coll, **T. Bukits**, R. Perisé and S. Cabrero Barros.

BAZKARIA: Eating Together Through Immersive Telepresence — EURO-XR 2025 · Poster
A. Santos-Torres, P. de Torres Coll, **T. Bukits**, A. Elosegui, I. Tamayo, R. Perisé, S. Cabrero Barros and M. Zorrilla.

Beyond Bounding Boxes: 2D Semantic Segmentation for Live Volumetric Video Streaming — ACM Multimedia 2025
T. Bukits, A. Elosegi, A. Dominguez and S. Cabrero Barros.

SKILLS

Research — 3D reconstruction, 3D Gaussian Splatting, neural rendering, volumetric capture, semantic segmentation, computer vision, computer graphics, deep learning, digital twins, extended reality

Programming — Python, PyTorch, TensorFlow/Keras, C/C++, CUDA, OpenGL, TypeScript, Node.js, Java

Languages — Hungarian (native), English (full professional proficiency), Spanish (full professional proficiency), German (basic)

AWARDS & RECOGNITION

Jun. 2026	Selected alumni presenter at IPCVAI Day 2026	Bordeaux, France
2023 – 2024	Two Erasmus+ scholarships supporting academic exchange semesters	Madrid & Bordeaux
2022 – 2024	Image Processing and Computer Vision Excellence Scholarship - being selected in a cohort of 12 students from more than 1,000 applicants worldwide	Budapest, Hungary
2017 – 2021	Integrated MSc Scholarship - selected for a high-achieving cohort and maintained a 4.0/5.0 GPA	Budapest, Hungary